Rocket Builder
BY QUIVER DESIGN WORKS

BIG KIT · 24 MM
What's in the tube

- A Nose cone
- B Top body section (cord strap inside)
- D Fin can with motor mount
- E Screw-on motor cap
- F Launch-lug standoff (1/8 in rod)
- G Elastic shock cord, 1/4 in, 5 ft
- H 15 in parachute, snap swivel
- I Recovery wadding, 6 sheets
- Waterproof decal sheet · quick-start card

Parts A–F are 3D-printed PLA, made in Ellensburg, WA. Letters match the assembly guide.

STEPS AT A GLANCE
Build it. Fly it.

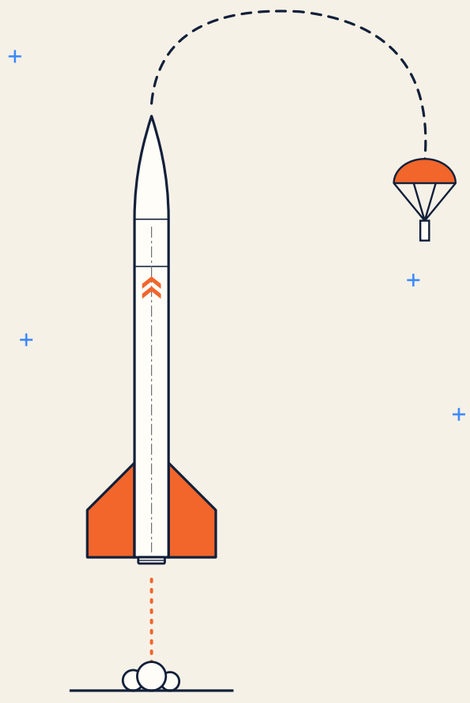
1. **Check the fit.** Slide A B D together. Snug, not forced.
2. **Tie the cord.** G through the strap in B, the other end to the nose A.
3. **Add the parachute.** Clip the swivel to the nose's tie bar. Hitch H to it.
4. **Glue and finish.** Glue B onto D, lugs in line. Add the decals.
5. **Motor in.** An adult puts in the motor and screws on cap E.
6. **Pack and launch.** Wadding I, parachute, cord, nose on. Count down from 5.

BIG KIT · 24 MM

Big rocket

A 3D-printed model rocket kit for D motors. Pre-printed parts, no painting. Comes in its own quiver.

SKILL 2 · ABOUT 1½ HOURS



FLIES ON 24 MM MOTORS

C11-5 **D12-7**

First flight: C11-5 (orange). Up to 420 m / 1,370 ft on a D12-7.

MOTORS NOT INCLUDED

Ages 12+ · Adults handle motors and launching.

You also need

24 mm motors: C11-5 or D12-7

Igniters and igniter plugs

Launch pad with a 1/8 in (3 mm) rod, and an electric launch controller

Super glue, scissors

Supply list: rocket.quiverdesignworks.net/ordersupplies

The rocket

Height	360 mm
Diameter	27 mm
Weight	71 g · 2.5 oz
Stability	1.1 cal
C11-5	200 m · 650 ft
D12-7	420 m · 1,370 ft

Where to fly

C motors	120 m · 400 ft
D motors	150 m · 500 ft

Smallest side of the field (NAR Model Rocket Safety Code). Get the landowner's OK.

FINS ARE FLETCHING
The fins keep the push point behind the balance point, like the feathers on an arrow. Design your own rocket free at **rocket.quiverdesignworks.net**.

SAFETY
Follow the NAR Model Rocket Safety Code. Launch outdoors in an open field, away from people, trees and power lines, in winds under 20 mph. Use a launch rod and an electric controller with a safety key. Stand back 5 m (15 ft). Never lean over the rocket. If a motor doesn't fire, pull the safety key and wait 60 seconds before going to the rocket.

QR
step-by-step guide

Full guide with pictures
Scan, or go to rocket.quiverdesignworks.net and open the guide.

Preflight checklist

- ☐ Cord tied at both ends
- ☐ Wadding first, chute packed loosely
- ☐ Motor cap E on all the way
- ☐ Igniter touches the propellant
- ☐ Safety key out until all are back
- ☐ Wind under 20 mph, no dry grass

WARNING: CHOKING HAZARD – Small parts. Not for children under 3 yrs.

For ages 12 and up. Launching needs adult supervision. Use only certified model rocket motors, installed by an adult.

Quiver Design Works LLC · Ellensburg, WA, USA · quiverdesignworks.net. Designed and 3D-printed in Ellensburg. Printed parts: PLA.
Batch **RB-B-_____** · Made __ / 20__ · Kit RB-KIT-M24

UPC-A placeholder